

Calculus I Lesson 23

1) Equation: $x^2 + y^2 = 9$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(x^2) = \underline{\quad? \quad} \cdot \frac{dx}{dt}$

b) $D_t(y^2) = \underline{\quad? \quad} \cdot \frac{dy}{dt}$

c) $D_t(9) = \underline{\quad? \quad}$

d) Write the implicit differential equation (with respect to t):

$$D_t(x^2) + D_t(y^2) = D_t(9) \quad \text{is} \quad \underline{\quad? \quad}$$

2) Equation: $\sqrt{x} + \sqrt{y} = 8$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(\sqrt{x}) = \underline{\quad? \quad} \cdot \frac{dx}{dt}$

b) $D_t(\sqrt{y}) = \underline{\quad? \quad} \cdot \frac{dy}{dt}$

c) $D_t(8) = \underline{\quad? \quad}$

d) Write the implicit differential equation (with respect to t):

$$D_t(\sqrt{x}) + D_t(\sqrt{y}) = D_t(8) \quad \text{is} \quad \underline{\quad? \quad}$$

3) Equation: $2x^3 - xy + y^2 = 12$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(2x^3) = \underline{\quad? \quad} \cdot \frac{dx}{dt}$

b) $D_t(-x \cdot y) = -x \cdot D_t(y) + y \cdot D_t(-x) = -x \cdot (1) \cdot \frac{dy}{dt} + y \cdot (-1) \cdot \frac{dx}{dt} = \underline{\quad? \quad}$

c) $D_t(y^2) = \underline{\quad? \quad}$

e) $D_t(12) = \underline{\quad? \quad}$

f) Write the implicit differential equation (with respect to t):

$$D_t(2x^3) + D_t(-x \cdot y) + D_t(y^2) = D_t(12) \quad \text{is} \quad \underline{\quad? \quad}$$

4) Equation: $4x^3 - 3x^2y + 2xy^2 = 10$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(4x^3) = \underline{\quad? \quad} \cdot \frac{dx}{dt}$

b) $D_t(-3x^2 \cdot y) = -3x^2 \cdot D_t(y) + y \cdot D_t(-3x^2) = -3x^2 \cdot (1) \cdot \frac{dy}{dt} + y \cdot (-6x) \cdot \frac{dx}{dt} = \underline{\quad? \quad}$

c) $D_t(2x \cdot y^2) = 2x \cdot D_t(y^2) + y^2 \cdot D_t(2x) = 2x \cdot (2y) \cdot \frac{dy}{dt} + y^2 \cdot (2) \cdot \frac{dx}{dt} = \underline{\quad? \quad}$

d) $D_t(10) = \underline{\quad? \quad}$

e) Write the implicit differential equation (with respect to t):

$D_t(4x^3) + D_t(-3x^2 \cdot y) + D_t(2x \cdot y^2) = D_t(10)$ is $\underline{\quad? \quad}$

5) Volume of right circular cylinder: $V = \pi \cdot r^2 \cdot h$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(V) = \underline{\quad? \quad} \cdot \frac{dV}{dt}$

b) $D_t(\pi \cdot r^2 \cdot h) = \pi \cdot r^2 \cdot D_t(h) + h \cdot D_t(\pi \cdot r^2) = \pi \cdot r^2 \cdot (1) \cdot \frac{dh}{dt} + h \cdot (2\pi r) \cdot \frac{dr}{dt} = \underline{\quad? \quad}$

c) Write the implicit differential equation (with respect to t):

$D_t(V) = D_t(\pi \cdot r^2 \cdot h)$ is $\underline{\quad? \quad}$

Note: $\frac{dV}{dt}$ is the rate of change of volume with respect to time.

$\frac{dh}{dt}$ is the rate of change of height with respect to time.

$\frac{dr}{dt}$ is the rate of change of radius with respect to time.

6) Volume of sphere: $V = \frac{4}{3}\pi \cdot r^3$

Differentiate both sides of the equation with respect to t (time).

a) $D_t(V) = \underline{\quad ? \quad} \cdot \frac{dV}{dt}$

b) $D_t\left(\frac{4}{3}\pi \cdot r^3\right) = \frac{4}{3}\pi \cdot D_t(r^3) = \frac{4}{3}\pi \cdot (\underline{\quad ? \quad}) \cdot \frac{dr}{dt}$

c) Write the implicit differential equation (with respect to t):

$$D_t(V) = D_t\left(\frac{4}{3}\pi \cdot r^3\right) \text{ is } \underline{\quad ? \quad}$$

Note: $\frac{dV}{dt}$ is the rate of change of volume with respect to time.

$\frac{dr}{dt}$ is the rate of change of radius with respect to time.

